switch is closed the two capacitors are put in parallel. After the motor has reached 80% full load speed, the centrifugal switch opens and running capacitor remains in the circuit. It is used to disconnect the starting capacitor from the supply after attaining almost 75% of full load speed.Either by changing the main winding terminals or by changing the starting winding terminals, the reversal of direction of rotation could be obtained.

APPLICATIONS

These motors are mainly used for low noise and high starting torque applications

1. Compressors
2. Pumps
3. Air Conditioners
4. Refrigerators.

c) Universal Motor

The universal motor is defined as a motor which may be operated on D.C supply or single phase A.C supply at approximately the same speed and output.

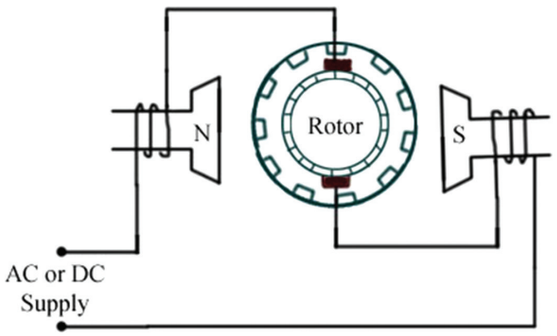


Fig 8. 8 *Universal Motor*

CONSTRUCTION

Basically universal motor is similar to D.C series motor. The universal motor is usually two poles and a winding of few turns which gives opposite magnetic polarity. The armature is of wound type and it consists of a laminated core having either straight (or) skewed slots and a commutator to which the leads of the armature winding are connected.

WORKING PRINCIPLE

Universal motor develop unidirectional torque, whether they operate on D.C or A.C. Supply Universal motor works on the same principle as a D.C motor, i.e. force between the main pole flux and the current carrying armature conductors.

Comparision of various types of single phase motors

Sl. No	Types of the Motor	Starting Torque	Speed	Windings	Applications
1	Split – phase motor	Low	Low	Main winding Starting winding	Driving fans, wood working tools
2	Capacitor start capacitor run motor	High	Constant	Main winding Starting winding	Compressors, refrigerators, air -Conditioners
3	Universal motor	Very high	Varies according to load	Field winding armature winding	Vacuum cleaners, mixie, portable drills, sewing machine

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